

MATTHEW GOODBRED

Department of Astrophysical Sciences, Princeton University ♦ 4 Ivy Lane, Princeton, NJ 08544

matthew.goodbred@gmail.com

EDUCATION

Princeton University, Princeton, NJ

September 2023 - Expected 2029

Ph.D. in Astrophysical Sciences

Dartmouth College, Hanover, NH

September 2019 - June 2023

B.A., Physics with High Honors (*Magna Cum Laude*); Minor in Mathematics

GPA: 3.94

Thesis: *The Structure and Rate of Relativistic Guide Field Magnetic Reconnection*

Phi Beta Kappa; Sigma Xi Scientific Research Honor Society

PUBLICATIONS

M. Goodbred and A. Spitkovsky, “Diocotron Modes in Pulsar Magnetospheres: Charge Diffusion and Implications for Radio Emission Variability”, *The Astrophysical Journal*, accepted (2026). arXiv:2606.03984

M. Goodbred and Y.-H. Liu, “First-Principles Theory of the Relativistic Magnetic Reconnection Rate in Astrophysical Pair Plasmas”, *Physical Review Letters* **129**, 265101 (2022)

M. Goodbred, Y.-H. Liu, B. Chen, and X. Li, “The relation between the energy conversion rate and reconnection rate in Petschek-type reconnection—Implications for solar flares”, *Physics of Plasmas* **28**, 082103 (2021)

RESEARCH EXPERIENCE

Graduate Research, Princeton University

September 2023 - Present

Advisor: Anatoly Spitkovsky

- Performing 3D particle-in-cell simulations (Tristan-MP v2) of aligned and oblique pulsar magnetospheres on national and institutional HPC resources, resolving the disk-dome equilibrium and the onset of the diocotron instability.
- Established that the nonlinear diocotron state is a long-lived $m = 1$ mode carrying a few percent of the central point charge, and derived a quasilinear radial diffusion coefficient for stochastic cross-field charge transport through the closed zone.
- Developed the multipole framework relating diocotron charge moments to polar-cap potential and emission-beam perturbations, connecting the instability to nulling, periodic amplitude modulation, and drifting subpulses.
- Wrote and maintain Python analysis and visualization pipelines for large 3D PIC datasets, plus an independent N-body electrosphere code for equilibrium studies.

Magnetic Reconnection Group, Dartmouth College

June 2020 - June 2023

Advisor: Yi-Hsin Liu

- Developed the first analytic theory for the rate of relativistic magnetic reconnection in magnetically dominated pair plasmas; validated predictions against large-scale PIC simulations (first-author *PRL*).
- Derived a theory relating energy conversion rate to reconnection rate in Petschek-type reconnection, with application to solar flares (first-author *Physics of Plasmas*).

Quantum Engineering Group, Dartmouth College

September 2022 - November 2022

Advisor: Sekhar Ramanathan

- Extended an AlphaZero-based reinforcement learning algorithm to design radio pulse sequences for quantum control, producing sequences outperforming the state of the art.

SELECTED TALKS AND PRESENTATIONS

A New Theory of Fast Magnetic Reconnection in Strongly Magnetized Plasmas

Oral, APS Division of Plasma Physics Meeting, Denver, CO

November 2023

The relation between the energy conversion rate and the reconnection rate in Petschek-type reconnection

- Oral, American Geophysical Union (AGU) Fall Meeting

December 2021

- Invited talk, SoLFER colloquium

October 2021

- Poster, SHINE Heliophysics 2050 conference

September 2021

AWARDS AND FELLOWSHIPS

Gazzaniga Family Science Award, Dartmouth College

2023

Awarded to the graduating senior with the strongest record of scientific research.

Francis W. Sears Physics and Astronomy Faculty Prize, Dartmouth College

2023

Awarded for significant departmental contributions and unusual undergraduate achievement.

Wilder Undergraduate Research Fellowship, Dartmouth College

2022

Francis L. Town Scientific Prize for Achievement in Physics, Dartmouth College

2021

James O. Freedman Presidential Scholarship, Dartmouth College

2021

TEACHING AND MENTORING

Assistant in Instruction, AST/PHY 401: Cosmology, Princeton University

Spring 2026

- Upper-level course on extragalactic astronomy and cosmology: galaxies, clusters and large-scale structure, dark matter and dark energy, the expanding and early Universe, the CMB, inflation, and structure formation.
- Led problem sessions and office hours, and graded problem sets and exams.

Learning Assistant, Physics 41 (Electrodynamics), Dartmouth College

Winter 2022

- Facilitated in-class problem sessions and weekly office hours; worked with the instructor on course design.

Physics and Computer Science Tutor, Dartmouth College

Winter 2021, Spring 2023

SERVICE AND OUTREACH

Physics/Astronomy Foreign Study Program, Cape Town, South Africa

January - March 2023

- Conducted observations at the South African Astronomical Observatory; participated in outreach to local schools.

MEDIA COVERAGE

The Tiny Physics Behind Immense Cosmic Eruptions, Z. Savitsky, *Quanta Magazine*

May 2023

Decoding Mega Magnetic Explosions Outside the Solar System, H. Barath, *Dartmouth News*

January 2023

TECHNICAL SKILLS

Simulation: PIC, large-scale HPC/MPI workflows, N-body electrostatic codes

Languages and Tools: Python (NumPy/SciPy/h5py/Matplotlib), Jupyter, MATLAB, Mathematica, C, $\text{\LaTeX}$, Git, Unix command line